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Part I

Basics

Chapter 1

Introduction

Signals are functions if independent variables that carry information. It can be continuous or discrete and be multi-dimensional. A system responds to applies input signals, and its response is described in terms of one or more output signals. A continuous time system receives and gives continuous signals and discrete time system receives and gives discrete signals. Systems can be connected together in series, parallel, or feedback loop.

1.1 Signal Properties

Definition: The energy of a continuous signal over interval $[t_1, t_2]$ is defined as

$$\int_{t_1}^{t_2} |x(t)|^2 \, \mathrm{d}t$$

and similarly for a discrete signal over interval $n_1 \leq n \leq n_2$

$$\sum_{n=n_1}^{n_2} |x[n]|^2$$

The power of signal is the time averaged energy of that signal. The total energy of signal is defined to be

$$E_\infty = \lim_{T\to\infty} \int_{-T} T|x(t)|^2 \, \mathrm{d}t$$

for a continuous time signal and

$$E_\infty = \lim_{N\to\infty} \sum_{n=-N}^{N} |x[n]|^2$$

Lastly, the total power of a signal is

$$P_\infty = \lim_{T\to\infty} \frac{1}{T} \int_{-T} T|x(t)|^2 \, \mathrm{d}t$$

and

$$P_\infty = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$$

1.2 System Properties

Memoryless A system is memoryless if it depends only on the present input. On the other hand, a system has memory if it depends on present and past values. For example, *accumulator* and *delay* are two such systems.

$$y[n] = \sum_{k=-\infty}^{n} x[n], \qquad y[n] = x[n-1]$$

Invertible A system is invertible if there exists a system $y(t) \to w(t)$ such that $w(t) = x(t)$, for all t.

Causal A system is casual or *nonancitipative* if it depends on past and present value. Mathematicall, a system $x(t) \to y(t)$ is causal if

$$x_1(t) = x_2(t), \quad \forall t \leq t_0 \implies y_1(t) = y_2(t), \quad \forall t \leq t_0$$

Stability Informally means that small change in the input does not converge. That is, bounded input results in bounded output.

Time invariant Time shift input results in an identical time shift in the output signal.

$$x(t) \to y(t) \implies x(t-t_0) \to y(t-t_0)$$

Linearity A system is linear if

$$ax_1(t) + x_2(t) \to ay_1(t) + y_2(t), \quad \forall a \in \mathbb{C}$$

Proposition 1.1. *A linear system is causal if and only if it satisfies the condition of initial rest*

$$x(t) = 0, \quad \forall t \leq t_0 \implies y(t) = 0, \quad \forall t \leq t_0$$

Proof. Consider a linear system such that the responses to $x_1(t)$ and $x_2(t)$ are $y_1(t)$ and $y_2(t)$, respectively. Suppos this system is causal. By linearity,$x_2(t) = 0 \implies y_2(t) = 0$ and hence

$$x_1(t) = x_2(t) = 0, \quad \forall t \leq t_0 \implies y_1(t) = y_2(t) = 0, \quad \forall t \leq t_0$$

Suppose the system has the initial rest condition. By linearity,

$$x_1(t) - x_2(t) \to y_1(t) - y_2(t)$$

Then, if $x_1(t) = x_2(t), \quad \forall t \leq t_0$ then

$$x_1(t) = x_2(t) \implies x_1(t) - x_2(t) = 0 \implies y_1(t) - y_2(t) = 0 \implies y_1(t) = y_2(t), \quad \forall t \leq t_0$$

Chapter 2

Linear Time Invariant Signals

2.1 Discrete signals

Let $x[n]$ be a discrete signal then it can be written as

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] = x[n] * \delta[n]$$

which is called *sifting property*. Consider a discrete LTI system $x[n] \to y[n]$, by sifting property

$$y[n] = x[n] * h[n]$$

where $h[n]$ is the response to $\delta[n]$. Hence a LTI system can be completely characterized by its response to $\delta[n]$.

2.2 Continuous signal

Similarly we can write the sifting propery as

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,\mathrm{d}\tau = x(t) * \delta(t)$$

and a continuous LTI system $x(t) \to y(t)$ can be written in terms of its response $h(t)$ to unit impluse $\delta(t)$

$$y(t) = x(t) * h(t)$$

2.3 Properties of convolution and LTI

For simplicity we only bring the continuous, however, the equivalent discrete form also holds.

Commutative $x(t) * y(t) = y(t) * x(t)$.

Distributive $x(t) * (y(t) + z(t)) = x(t) * y(t) + x(t) * z(t)$.

Associative $x(t) * (y(t) * z(t)) = (x(t) * y(t)) * z(t)$.

- LTI is memoryless if $h(t) = 0$ for $t \neq 0 \implies h(t) = K\delta(t) \implies y(t) = Kx(t)$.

- LTI system is invertible if there exists $g(t)$ such that $h(t) * g(t) = \delta(t)$.
- LTI system is causal if $h(t) = 0$ for $t < 0$.
- LTI is stable iff
$$\int_{-\infty}^{\infty} |h(\tau)| \, \mathrm{d}\tau < \infty$$

2.4 Singularity functions

We can view the unit impluse as a short signal that has integral of 1 and hence signals with such property are similar. Consider the LTI system

$$y(t) = \frac{\mathrm{d}^n x(t)}{\mathrm{d}t^n}$$

Then the unit impluse response is $u_n(t)$ such that

$$y(t) = x(t) * u_n(t) = (x(t) * u_{n-1}(t)) * u_1(t)$$

and therefore

$$u_n(t) = \underbrace{u_1(t) * u_1(t) * \cdots * u_1(t)}_{n}$$

$u_1(t)$ is called the unit doublet and it is defined

$$u_1(t) = \frac{\mathrm{d}\delta(t)}{\mathrm{d}t}$$

Similarly for integral

$$u_{-1}(t) = u(t) = \int_{-\infty}^{t} \delta(\tau) \, \mathrm{d}\tau \implies x(t) * u(t) = \int_{-\infty}^{t} x(\tau) \, \mathrm{d}\tau$$

for the double integral

$$u_{-2}(t) = u_{-1}(t) u_{-1}(t) = \int_{-\infty}^{t} u_{-1}(\tau) \, \mathrm{d}\tau$$

similarly for higher order integrals

$$u_{-n}(t) = \underbrace{u_{-1}(t) * u_{-1}(t) * \cdots * u_{-1}(t)}_{n}$$

Lastly we can denote $\delta(t) = u_0(t)$ to then arrive at

$$u_r(t) * u_s(t) = u_{r+s}(t)$$

for all $u, s \in \mathbb{Z}$.

Part II

Fourier

Chapter 3

Fourier Series Representation

3.1 Eigenfunctions

For discrete and continuous LTI systems we have

$$x(t) = e^{st} \to y(t) = H(s)e^{st},\ s \in \mathbb{C}$$
$$x[n] = z^n \to y[n] = H(x)z^n,\ z \in \mathbb{C}$$

A signal for which the output signal of a system is a constant multiple of itself is called the **eigenfunction** of the system and the constant is called the **eigenvalue**. To show the above equations (we assume the convergence of the integral and the sum at the last steps.)

$$y(t) = \int_{-\infty}^{\infty} h(\tau)e^{s(t-\tau)}\,\mathrm{d}\tau = e^{st}\int_{-\infty}^{\infty} h(\tau)e^{-s\tau}\,\mathrm{d}\tau = H(s)e^{st}$$
$$y[n] = \sum_{k=-\infty}^{\infty} h[k]z^{n-k} = z^n \sum_{k=-\infty}^{\infty} h[k]z^{-k} = H(z)z^n$$

3.2 Fourier series representation

3.2.1 Continuous-time

Suppose $x(t)$ is a continuous-time periodic signal with period T. The value $\omega_0 = \frac{2\pi}{T}$ then we may be able to write $x(t)$ in the form of

$$x(t) = \sum_{-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

Let $x(t)$ be a real signal then $\overline{x} = x$ hence

$$\begin{aligned} x(t) &= \sum_{-\infty}^{\infty} \overline{a_k} e^{-jk\omega_0 t} \\ &= \sum_{-\infty}^{\infty} \overline{a_{-k}} e^{-jk\omega_0 t} \implies a_k = \overline{a_{-k}} \\ &= a_0 + \sum_{k=1}^{\infty} a_k e^{jk\omega_0 t} + \overline{a_{-k}} e^{-jk\omega_0 t} \end{aligned}$$

$$
\begin{aligned}
&= a_0 + 2\sum_{k=1}^{\infty} \Re\big(a_k e^{jk\omega_0 t}\big),\ a_k = r_k e^{j\theta_k}, a_k = b_k + jc_k \\
&= a_0 + 2\sum_{k=1}^{\infty} r_k \cos(\theta_k + k\omega_0 t) \\
&= a_0 + 2\sum_{k=1}^{\infty} b_k \cos(k\omega_0 t) - c_k \sin(k\omega_0 t)
\end{aligned}
$$

The last two equation are the **Fourier series** representation of a real periodic signal $x(t)$ _ this is called synthesis in the context of Fourier series. We also have

$$
\begin{aligned}
\int_0^T x(t)e^{-jn\omega_0 t}\,\mathrm{d}t &= \int_0^T \sum_{-\infty}^{\infty} a_k e^{j(n-k)\omega_0 t}\,\mathrm{d}t \\
&= \sum_{-\infty}^{\infty} a_k \int_0^T e^{j(n-k)\omega_0 t}\,\mathrm{d}t \\
&= a_n T
\end{aligned}
$$

therefore

$$
\begin{aligned}
a_n &= \frac{\omega_0}{2\pi}\int_0^T x(t)e^{-jn\omega_0 t}\,\mathrm{d}t = \frac{\omega_0}{2\pi}\int_c^{c+T} x(t)e^{-jn\omega_0 t}\,\mathrm{d}t \\
a_0 &= \frac{\omega_0}{2\pi}\int_0^T x(t)\,\mathrm{d}t
\end{aligned}
$$

we donote $\int_c^{c+T}$ by $\int_T$.

3.2.2 Convergence

A finite enery real signal $x(t)$

$$\int_T |x(t)|^2\,\mathrm{d}t < \infty$$

has a Fourier series representation if it satisfies the *Dirichlet conditions*

1. Over any period $x(t)$ is absolutely intergrable which implies $|a_k| < \infty$.
2. In any finite interval, $x(t)$ is of bounded variation _ finite number of maximas and minimas.
3. Finite number of discontinuities in any finite interval, and the discontinuities can not be infinite.

3.2.3 Discrete-time

Let $x[n]$ be a discrete signal with period N and fundamental frequence $w_0 = \frac{2\pi}{N}$ and let

$$\phi_k(n) = e^{jk\omega_0 n} \quad k \in \mathbb{Z}$$

then since

$$\phi_k(n) = \phi_{rN+k}(n)$$

we can write $x[n]$ in form of

$$x(n) = \sum_{k=\langle N\rangle} a_k \phi_k(n)$$

where $\langle N\rangle$ is a set of N consecutive integers. Finding a_k requires solving a linear system

$$a_k = \frac{1}{N} \sum_{k=\langle N\rangle} x[n] e^{jk\omega_0 n}$$

3.3 Properties of Fourier series

Suppose $x(t)$ is a periodic signal with period T and $\omega_0 = \frac{2\pi}{T}$ then

$$x(t) \overset{FS}{\longleftrightarrow} a_k$$

where a_k is its Fourier series coefficients. Let $x(t) \overset{FS}{\longleftrightarrow} a_k$ and $y(t) \overset{FS}{\longleftrightarrow} b_k$, then we have the following properties

Definition:

Linearity

$$\alpha x(t) + \beta y(t) \overset{FS}{\longleftrightarrow} \alpha a_k + \beta b_k$$

Time shifting

$$x(t - t_0) \overset{FS}{\longleftrightarrow} e^{-jk\omega_0 t_0} a_k$$

Time reversal

$$\alpha x(-t) \overset{FS}{\longleftrightarrow} a_{-k}$$

Time scaling

$$\alpha x(ct) \overset{FS}{\longleftrightarrow} \sum_{k=-\infty}^{\infty} a_k e^{jk(c\omega_0)t}$$

It does not change the coefficients, but changes the whole thing.

Multiplication

$$x(t)y(t) \overset{FS}{\longleftrightarrow} c_k = \sum_{l=-\infty}^{\infty} a_l + b_{k-l}$$

Conjugate symmetry

$$\overline{x(t)} \overset{FS}{\longleftrightarrow} \overline{a_{-k}}$$

Parseval's relation

$$\int_T |x(t)|^2 \,\mathrm{d}t = \sum_{k=-\infty}^{\infty} |a_k|^2$$

Period convolution

$$\int_T x(t)y(t-\tau)\,\mathrm{d}\tau \overset{FS}{\longleftrightarrow} T\alpha a_k \beta b_k$$

Frequency shifting

$$e^{jM\omega_0 t}x(t) \overset{FS}{\longleftrightarrow} a_{k-M}$$

Differentiation

$$\frac{\mathrm{d}x(t)}{\mathrm{d}t} \overset{FS}{\longleftrightarrow} jk\omega_0 \alpha a_k$$

and for discrete-time signals

$$x[n] - x[n-1] \overset{FS}{\longleftrightarrow} \left(1 - e^{-jk\omega_0}\right) a_K$$

Integeration

$$\int_{\infty}^{t} x(\tau)\,\mathrm{d}\tau \overset{FS}{\longleftrightarrow} \frac{a_k}{jk\omega_0}$$

3.4 LTI systems

Consider a periodic signal $x(t)$ with

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

then for a LTI system $x(t) \to y(t)$ with $\delta(t) \to h(t)$

$$\begin{aligned}
y(t) &= \sum_{k=-\infty}^{\infty} a_k \left(e^{jk\omega_0 t} * h(t)\right) \\
&= \sum_{k=-\infty}^{\infty} a_k \left(\int_{-\infty}^{\infty} e^{jk\omega_0 (t-\tau)} h(\tau)\,\mathrm{d}\tau\right) \\
&= \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \left(\int_{-\infty}^{\infty} e^{-jk\omega_0 \tau} h(\tau)\,\mathrm{d}\tau\right) \\
&= \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) e^{jk\omega_0 t}
\end{aligned}$$

For a discrete time system $\delta[n] \to h[n]$

$$\begin{aligned}
y(n) &= \sum_{k=\langle N\rangle} a_K \left(e^{jk\omega_0 n} * h(n)\right) \\
&= \sum_{k=\langle N\rangle} a_K \left(\left(e^{jk\omega_0}\right)^n * h(n)\right) \\
&= \sum_{k=\langle N\rangle} a_K \left(\sum_{m=-\infty}^{\infty} e^{jk\omega_0 (n-m)} h(m)\right) \\
&= \sum_{k=\langle N\rangle} a_K e^{jk\omega_0 n} \left(\sum_{m=-\infty}^{\infty} e^{-jk\omega_0 m} h(m)\right) \\
&= \sum_{k=\langle N\rangle} a_K H\left(e^{jk\omega_0}\right) e^{jk\omega_0 n}
\end{aligned}$$

3.5 Filtering

LTI systems the change frequency spectrum are called frequency-shaping filters. Systems that pass, eliminate, or attunate some frequencies are called frequency selective filters. Three types of frequency selective filters include

3.5.1 Lowpass filter

Passes low frequencies around zero and attunates or eliminates, in the ideal case, high frequencies.

$$|H|j\omega = \begin{cases} 1 & |\omega| \leq \omega_c \\ 0 & \text{otherwise} \end{cases}$$

–insert diagram

3.5.2 Bandpass filter

Passes frequencies around certian frequncy and attunates or eliminates, in the ideal case, other frequencies.

$$|H|j\omega = \begin{cases} 1 & \omega_{c_1} \leq |\omega| \leq \omega_{c_2} \\ 0 & \text{otherwise} \end{cases}$$

–insert diagram

3.5.3 High pass filter

Passes high frequencies around zero and attunates or eliminates, in the ideal case, low frequencies.

$$|H|j\omega = \begin{cases} 1 & |\omega| \geq \omega_c \\ 0 & \text{otherwise} \end{cases}$$

–insert diagram

Example 3.1. Take a simple RC circuit with input $V_s(t)$. We have

$$\begin{cases} V_R(t) + V_C(t) = V_s(t) \\ V_R(t) = Ri(t) = RC\frac{\mathrm{d}V_C(t)}{\mathrm{d}t} \end{cases}$$

Since it is an LTI system, for input $V_s(t) = e^{j\omega t}$ and output $V_C(t) = H(j\omega)$ we have

$$RCH(j\omega)j\omega e^{j\omega t} + H(j\omega)e^{j\omega} = e^{j\omega}$$
$$\implies H(j\omega) = \frac{1}{RCj\omega + 1}$$

–insert diagram which is non-ideal lowpass filter. For output $V_R(t) = H(j\omega)$ we have

$$RCH(j\omega)j\omega e^{j\omega t} + H(j\omega)e^{j\omega} = RCj\omega e^{j\omega}$$
$$\implies H(j\omega) = \frac{j\omega}{RCj\omega + 1}$$

–insert diagram which is non-ideal highpass filter.

Example 3.2 (First order recursive discrete-time filter). Consider

$$y[n] - ay[n-1] = x[n]$$

with input $x[n] = e^{j\omega n}$ the output is of form $y[n] = H(e^{j\omega})e^{j\omega n}$ then

$$\begin{aligned} H\left(e^{j\omega}\right)e^{j\omega n} + H\left(e^{j\omega}\right)e^{j\omega(n-1)} &= e^{j\omega n} \\ \implies H\left(e^{j\omega}\right)\left(1 + e^{-j\omega}\right) &= 1 \\ \implies H\left(e^{j\omega}\right) &= \frac{1}{1 + e^{-j\omega}} \end{aligned}$$

–insert diagram which is non-ideal lowpass filter for $0 < a < 1$ and a non-ideal highpass filter for $-1 < a < 0$.

Example 3.3 (Non-recursive discrete-time filter). Consider the following FIR

$$y[n] = \sum_{k=-N}^{M} b_k x[n-k], \quad b_k \text{ are weights}$$

let $b_k = \frac{1}{M+N+1}$ then

$$\begin{aligned} H\left(e^{j\omega}\right) &= \frac{1}{N+M+1} \sum_{k=-N}^{M} e^{j\omega(n-k)} \\ &= \frac{1}{N+M+1} \frac{e^{j\omega(N+1)} - e^{-j\omega M}}{e^{j\omega} - 1} \end{aligned}$$

which is a lowpass filter that approaches the ideal as $N + M + 1 \to \infty$.

Chapter 4

Continuous Fourier Transform

4.1 Fourier transform for a periodic function

Suppose $x(t)$ with $x(t) = 0$ for $|t| \geq T_1$ is a finite duration aperiodic signal. Make a periodic signal $\tilde{x}(t)$ a from $x(t)$ with period $T \geq T_1$. Then $\tilde{x}$ has a Fourier series

$$\begin{aligned}
\tilde{x}(t) &= \sum_k a_k e^{jk\omega_0 t} \qquad \omega_0 = \frac{2\pi}{T} \\
a_k &= \frac{1}{T}\int_T \tilde{x}(t) e^{-jk\omega_0 t}\,\mathrm{d}t \\
&= \frac{1}{T}\int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t}\,\mathrm{d}t \\
&= \frac{1}{T}\int_{-\infty}^{\infty} x(t) e^{-jk\omega_0}\,\mathrm{d}t \\
&= \frac{1}{T} X(jk\omega_0)
\end{aligned}$$

since for $|t| \leq T/2, x(t) = \tilde{x}(t)$ and for $|t| > T \geq T_1$, $x(t) = 0$. $X(j\omega)$ is the *Fourier integral* or *Fourier transform* of x.

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\,\mathrm{d}t$$

Therefore,

$$\begin{aligned}
\tilde{x}(t) &= \sum_k \frac{1}{T} X(jk\omega_0) e^{jk\omega_0 t} \\
&= \frac{1}{2\pi}\sum_k X(jk\omega_0) e^{jk\omega_0 t}\omega_0
\end{aligned}$$

as $T \to \infty$, $\tilde{x} \to x$ and $\omega_0 \to 0$ hence

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega) e^{j\omega t}\,\mathrm{d}\omega$$

For this to converge we need the Dirichlet condition, that is $x(t)$ must be

1. absolutely integrable.

2. bounded variation.

3. finite number of finite discontinuities.

For periodic signals the Fourier transform is a collection of impulse signals occuring at harmonic

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}\,\mathrm{d}\omega$$

Let $X(j\omega) = 2\pi\sum_{k=-\infty}^{\infty} a_k\delta(\omega - k\omega_0)$

$$\begin{aligned}
&= \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(2\pi\sum_{k=-\infty}^{\infty} a_k\delta(\omega - k\omega_0)\right)e^{j\omega t}\,\mathrm{d}\omega \\
&= \sum_{k=-\infty}^{\infty} a_k\int_{-\infty}^{\infty}\delta(\omega - k\omega_0)e^{j\omega t}\,\mathrm{d}\omega \\
&= \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}
\end{aligned}$$

or equivalently

$$\begin{aligned}
X(j\omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,\mathrm{d}t \\
&= \sum_{k=-\infty}^{\infty} a_k\int_{-\infty}^{\infty} e^{jk\omega_0 t}e^{-j\omega t}\,\mathrm{d}t \qquad \text{(distribution integra)} \\
&= \sum_{k=-\infty}^{\infty} a_k\delta(\omega - k\omega_0)
\end{aligned}$$

4.2 Properties

Linearity

$$\mathcal{F}\{ax(t) + by(t)\} = a\mathcal{F}\{x(t)\} + b\mathcal{F}\{y(t)\}$$

Time shifting

$$\mathcal{F}\{x(t - t_0)\} = e^{-j\omega t_0}\mathcal{F}\{x(t)\}$$

Conjugate symmetries

$$\mathcal{F}\left\{\overline{x(t)}\right\} = \overline{\mathcal{F}\{x(-t)\}}$$

Therefore, for a real signal x

$$\begin{aligned}
&\mathcal{F}\{x(-t)\} = \overline{\mathcal{F}\{x(t)\}} \\
\implies &\mathcal{F}\{Od(x(t))\} = \frac{\mathcal{F}\{x(t)\} - \overline{\mathcal{F}\{x(t)\}}}{2} = i\Im\mathcal{F}\{x(t)\} \\
\implies &\mathcal{F}\{Ev(x(t))\} = \frac{\mathcal{F}\{x(t)\} + \overline{\mathcal{F}\{x(t)\}}}{2} = \Re\mathcal{F}\{x(t)\}
\end{aligned}$$

Derivatives

$$\mathcal{F}\left\{\frac{\mathrm{d}^n}{\mathrm{d}t^n}x(t)\right\} = (j\omega)^n\mathcal{F}\{x(t)\}$$

Convolution

$$\mathcal{F}\{x(t) * y(t)\} = \mathcal{F}\{x(t)\}\mathcal{F}\{x(t)\}$$

Time scaling

$$\mathcal{F}\{x(at)\} = \frac{1}{|a|}\mathcal{F}\{x(t)\}\left(\frac{\omega}{a}\right)$$

Integral

$$\mathcal{F}\left\{\int_{\infty}^{t} x(\tau)\,\mathrm{d}\tau\right\} = \frac{1}{j\omega}\mathcal{F}\{x(t)\} + \pi\mathcal{F}\{x(t)\}(0)\delta(\omega)$$

Inverse of derivative

$$\mathcal{F}^{-1}\left\{\frac{\mathrm{d}}{\mathrm{d}\omega}\mathcal{F}\{x(t)\}\right\} = -jtx(t)$$

Inverse of frequency shifting

$$\mathcal{F}^{-1}\{\mathcal{F}\{x(t)\}(\omega - \omega_0)\} = e^{j\omega_0 t}x(t)$$

Multiplication

$$\mathcal{F}\{x(t)y(t)\} = \mathcal{F}\{x(t)\} * \mathcal{F}\{y(t)\}$$

Parseval

$$\int_{\infty}^{\infty} |x(t)|^2\,\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} |\mathcal{F}\{x(t)\}|^2\,\mathrm{d}\omega$$

Fourier transform of some functions

$x(t)$	$\mathcal{F}\{x(t)\}$
1	$2\pi\delta(\omega)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos(\omega_0 t)$	$\pi(\delta(\omega - \omega_0) + \delta(\omega + \omega_0))$
$\sin(\omega_0 t)$	$\frac{\pi}{j}(\delta(\omega - \omega_0) - \delta(\omega + \omega_0))$
$\delta(t)$	1
$\frac{t^{n-1}}{(n-1)!}e^{-at}u(t)$	$\frac{1}{(a + j\omega)^n} \quad \Re a > 0$

4.3 Applications

We can make frequency-selective filtering with variable center frequency (bandpass around ω_c. Solve ODEs in the following format

$$\sum_{k=0}^{K} a_k \frac{\mathrm{d}^k}{\mathrm{d}t^k} y(t) = \sum_{m=0}^{M} a_m \frac{\mathrm{d}^m}{\mathrm{d}t^m} x(t)$$

since it is an LTI system then $y(t) = x(t) * h(t)$ and

$$Y(j\omega) = \mathcal{F}\{y(t)\} = \mathcal{F}\{x(t) * h(t)\} = H(j\omega)X(j\omega)$$

therefore

$$H(j\omega) = \frac{\sum_{m=0}^{M} b_k (j\omega)^m}{\sum_{k=0}^{K} a_k (j\omega)^k}$$

Chapter 5

Discrete Fourier Transform

Let $x[n]$ be a finite duration signal and $\tilde{x}[n]$ be the periodic form of $x[n]$, Then, as $N \to \infty$ $\tilde{x}[n] \to x[n]$.

$$\begin{aligned}
\tilde{x}[n] &= \sum_{\langle N \rangle} a_k e^{j\omega_0 kn} \qquad \omega_0 = \frac{2\pi}{n} \\
\implies a_k &= \frac{1}{N} \sum_{\langle N \rangle} \tilde{x}[n] e^{-j\omega_0 kn} \\
&= \frac{1}{N} \sum_{n=-N_1}^{N_2} x[n] e^{-j\omega_0 kn} \\
&= \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega_0 kn} \\
&= \frac{1}{N} X\left(e^{j\omega_0 k}\right)
\end{aligned}$$

which implies that

$$\begin{aligned}
\tilde{x}[n] &= \sum_{\langle N \rangle} \frac{1}{N} X\left(e^{j\omega_0 k}\right) e^{j\omega_0 kn} \\
\implies x[n] &= \lim_{\omega_0 \to \infty} \frac{1}{2\pi} \sum_{0}^{\frac{2\pi}{n}} X\left(e^{j\omega_0 k}\right) e^{j\omega_0 kn} \omega_0 \\
&= \frac{1}{2\pi} \int_{2\pi} X\left(e^{j\omega}\right) e^{j\omega n} \, \mathrm{d}\omega
\end{aligned}$$

For convergence we need the followings to hold

1. $x[n]$ is absolutely summable

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

   or has finite energy

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty$$

5.1 DFT of periodic signal

Let $x[n] = e^{j\omega_0 n}$ then

$$\begin{aligned} X\left(e^{j\omega}\right) &= \sum_{n=-\infty}^{\infty} e^{j\omega_0 n} e^{-j\omega n} = 2\pi \sum_{l=-\infty}^{\infty} \delta(\omega - \omega_0 - 2\pi l) \\ \implies x[n] &= \sum_{\langle N \rangle} a_k e^{jk\omega_0 n} \\ \implies X\left(e^{j\omega}\right) &= 2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_0) \end{aligned}$$

5.2 Properties

Periodic It has a period of 2π.

$$\mathcal{F}\{x[n]\}(\omega + 2\pi) = \mathcal{F}\{x[n]\}(\omega)$$

Linear

$$\mathcal{F}\{ax[n] + by[n]\} = a\mathcal{F}\{x[n]\} + b\mathcal{F}\{y[n]\}$$

Time shifting n_0 must be an integer.

$$\mathcal{F}\{x[n - n_0]\} = e^{-j\omega n_0}\mathcal{F}\{x[n]\}$$

Frequency shifting

$$\mathcal{F}\left\{e^{j\omega_0 n} x[n]\right\} = \mathcal{F}\{x[n]\}(\omega - \omega_0)$$

Conjugate symmetry

$$\mathcal{F}\left\{\overline{x[n]}\right\} = \overline{\mathcal{F}\{x[n]\}(-\omega)}$$

and if $x[n]$ is real

$$\begin{aligned} \implies \mathcal{F}\{x[n]\} &= \overline{\mathcal{F}\{x[n]\}(-\omega)} \\ \implies \mathcal{F}\{Ev(x[n])\} &= \Re\mathcal{F}\{x[n]\} \\ \implies \mathcal{F}\{Od(x[n])\} &= j\Im\mathcal{F}\{x[n]\} \end{aligned}$$

Differencing

$$\mathcal{F}\{x[n] - x[n-1]\} = \left(1 - e^{-j\omega}\right)\mathcal{F}\{x[n]\}$$

Accumulation

$$\mathcal{F}\left\{\sum_{m=-\infty}^{n} x[m]\right\} = \frac{1}{1 - e^{-j\omega}}\mathcal{F}\{x[n]\} + \pi\mathcal{F}\{x[n]\}(0) \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

Time Rerversal

$$\mathcal{F}\{x[-n]\} = \mathcal{F}\{x[n]\}(-\omega)$$

Time Expansion

$$\mathcal{F}\left\{x_{(k)}[n]\right\} = \mathcal{F}\{x[n]\}(k\omega)$$

where

$$x_{(k)}[n] = \begin{cases} x\left[\frac{n}{k}\right] & k|n \\ 0 & \text{otherwise} \end{cases}$$

$$\mathcal{F}^{-1}\left\{j\frac{\mathrm{d}\mathcal{F}\{x[n]\}}{\mathrm{d}\omega}\right\} = nx[n]$$

Parsevals'

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{2\pi} |\mathcal{F}\{x[n]\}|^2 \,\mathrm{d}\omega$$

Convolution

$$\mathcal{F}\{x[n] * y[n]\} = \mathcal{F}\{x[n]\}\mathcal{F}\{y[n]\}$$

Multiplication

$$\mathcal{F}\{x[n]y[n]\} = \frac{1}{2\pi}\int_{2\pi} \mathcal{F}\{x[n]\}(\theta)\mathcal{F}\{y[n]\}(\omega-\theta)\,\mathrm{d}\theta$$

5.3 Linear constant coefficient difference equation

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{m=0}^{M} b_m x[n-m]$$

it is the impluse response has the following form

$$H\left(e^{j\omega}\right) = \frac{\sum_{m=0}^{M} b_m e^{-j\omega m}}{\sum_{k=0}^{N} a_k e^{j\omega k}}$$

Some fourier analysis would not be bad :)

Chapter 6

Time Frequency Characterization

6.1 Magnitude-phase representation of Fourier Transform

$\angle X(j\omega)$ is the relative phase and $\frac{1}{2\pi}|X(j\omega)|^2\,\mathrm{d}\omega$ is the energy of $x(t)$ in $]\omega, \omega+\mathrm{d}\omega[$. Obviously, the magnitude matters but also phase matters depending on the context.

6.2 Magnitude-phase representation of the frequency response of LTI systems

In an LTI system

$$Y(j\omega) = H(j\omega)X(j\omega) \implies \begin{cases} |Y| & = |H||X| \\ \angle Y & = \angle H + \angle X \end{cases}$$

where the $|H|$ is called the gain and $\angle H$ is the phase shift.

Linear and non-linear phase

For unit gain all-pass $H(j\omega) = 1$ with linear phase $\angle H(j\omega) = t_0\omega$. Then,

$$y(t) = x(t - t_0)$$

Group delay

Suppose $X(j\omega)$ is zero outside a narrow band around $\omega = \omega_0$. Then, a non-linear phase can be approximated by a linear phase

$$\begin{aligned} \angle H(j\omega) &\simeq -\phi - \alpha \\ \implies Y(j\omega) &= X(j\omega)|H(j\omega)|e^{j\phi}e^{j\omega\alpha} \end{aligned}$$

α seconds delay. The group delay is defined as

$$\tau(\omega) = -\frac{\mathrm{d}}{\mathrm{d}\omega}\angle H(j\omega)$$

but this might be discontinuous at $2\pi k$. So we use the un-wrapped phase

$$\tau(\omega) = -\frac{\mathrm{d}}{\mathrm{d}\omega}[\angle H(j\omega)]$$

Log-magnitude and Bode plots

$dB = 20\log_{10}|H(j\omega)|$. If ω-axis is logarithmic $\log_{10}\omega$ as well then it is called Bode plot.

6.3 Time-Domain properties of ideal frequency selective filter

a frequency selective filter

$$|H(j\omega)| = \begin{cases} 1 & |\omega| \leq \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$$

With linear phase (α seconds delay)

$$\angle H(j\omega) = -\alpha\omega$$

6.4 Time-domain and frequency aspects of non-ideal filters

add images in pg 175

6.5 First and second order CT system

First order continuous-time system

$$\tau\frac{\mathrm{d}y}{\mathrm{d}t} + u = x(t) \implies H(j\omega) = \frac{1}{j\omega\tau + 1}$$

which then implies that

$$\begin{aligned} h(t) &= \frac{1}{\tau}e^{-\frac{t}{\tau}}u(t) \\ s(t) &= \left(1 - e^{-\frac{t}{\tau}}\right)u(t) \end{aligned}$$

Second order continuous-time system

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\xi\omega_n\frac{\mathrm{d}y}{\mathrm{d}t} + \omega_n^2 y(t) = \omega_n^2 x(t)$$

then

$$\begin{aligned} H(j\omega) &= \frac{\omega_n^2}{(j\omega)^2 + 2\xi\omega_n j\omega + \omega_n^2} \\ &= \frac{\omega_n^2}{(j\omega - c_1)(j\omega - c_2)} \end{aligned}$$

where $c_1 = -\xi\omega_n + \omega_n\sqrt{\xi^2 - 1}$ and $c_2 = -\xi\omega_n + \omega_n\sqrt{\xi^2 - 1}$. If $\xi \neq 1$ then

$$H(j\omega) = \frac{M}{j\omega - c_1} - \frac{M}{j\omega - c_2} \qquad M = \frac{\omega_n}{2\sqrt{\xi^2 - 1}}$$

$$\implies h(t) = M\left(e^{c_1 t} - e^{c_2 t}\right)u(t)$$

and if $\xi = 1$ then

$$H(j\omega) = \frac{\omega_n^2}{(j\omega + \omega_n)^2} \implies h(t) = \omega_n^2 t e^{-\omega_n t} u(t)$$

ξ is called the damping ratio and ω_n is the undamped natural frequency.

- For $0 < \xi < 1$ the response is underdamped which will overshoot and rings in step function.
- For $\xi = 1$ the response is critically damped which will have the fastest settling time.
- For $\xi > 1$ the response is overdamped which will imply it has slow settling time.

6.6 First order and second order discrete-time system

6.7 First order

$$\begin{aligned} y[n] - ay[n-1] = x[n] \implies H\left(e^{j\omega}\right) &= \frac{1}{1 - ae^{j\omega}} \\ h[n] &= a^n u(n) \\ s[n] &= \frac{1 - a^{n+1}}{1 - a} u(n) \end{aligned}$$

6.8 Second order

$$y[n] - 2r\cos\theta y[n-1] + r^2 y[n-2] = x[n]$$

$$\begin{aligned} H\left(e^{j\omega}\right) &= \frac{1}{1 - 2r\cos\theta e^{-j\omega} + r^2 e^{-2j\omega}} \\ &= \frac{1}{(1 - re^{j\theta}e^{-j\omega})(1 - re^{=j\theta}e^{-j\omega})} \end{aligned}$$

If $\theta \neq 0, \pi$ then

$$H\left(e^{j\omega}\right) = \frac{A}{1 - re^{j\theta}e^{-j\omega}} + \frac{B}{1 - re^{-j\theta}e^{-j\omega}}$$

where

$$A = \frac{e^{j\theta}}{2j\sin\theta} \qquad B = \frac{e^{-j\theta}}{2j\sin\theta}$$

if $\theta = 0$

$$H\left(e^{j\omega}\right) = \frac{1}{(1 - re^{-j\omega})^2} \implies h[n] = (n+1)r^n u[n]$$

and for $\theta = \pi$

$$H\left(e^{j\omega}\right) = \frac{1}{(1 + re^{-j\omega})^2} \implies h[n] = (n+1)(-r)^n u[n]$$

Chapter 7

Sampling

7.1 Sampling theorem

Sampling with impluse train

Let T be the sampling period and $\omega_s = \frac{2\pi}{T}$ be the sampling frequency. The train impluse

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

and hence the sampled signal

$$x_p(t) = x(t)p(t) = \sum_{n=-\infty}^{\infty} x(nT)\delta(t - nT)$$

has the Fourier transform

$$X_p(j\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\theta)P(j\omega - j\theta)\,\mathrm{d}\theta = \frac{1}{2\pi} X(j\omega) * P(j\omega)$$

with

$$P(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s)$$

and hence

$$\begin{aligned} X_p(j\omega) &= \frac{\omega_s}{2\pi} X(j\omega) * \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s) \\ &= \frac{\omega_s}{2\pi} \sum_{k=-\infty}^{\infty} X(j\omega - jk\omega_s) \end{aligned}$$

Theorem 7.1. *Let $x(t)$ be a band-limited signal with $X(j\omega) = 0$ for $|\omega| > \omega_M$. Then, $x(t)$ is uniquely determined by its samples $x(nT)$ if $\omega_s > 2\omega_M$, i.e. we can reconstructed $x(t)$ from its samples.*

Proof. Construct $X_p(j\omega)$ as above. Note that is has a period of ω_s. Therefore, by the fact that $X(j\omega)$ is band-limited, if $\omega_M < \omega_s - \omega_M$ or equivalently $\omega_s > 2\omega_M$ then we can reconstruct $X(j\omega)$ by applying a lowpass filter with $\omega_M < \omega_c < \omega_s - \omega_M$. ■

Definition: $2\omega_M$ is called the Nyquist rate.

Sampling with zero-order hold

it is the same as impulse train by we hold the last sample until the new sample. – add images pg 178 where

$$H_0(j\omega) = e^{-j\omega T/2}\left(\frac{2\sin\frac{\omega T}{2}}{\omega}\right)$$

and therfore to have $x(t) = r(t)$ we must have

$$H_r(j\omega) = \frac{H(j\omega)}{H_0(j\omega)}$$

where $H(j\omega)$ is the ideal lowpass needed to convert impulse train to original signal.

7.2 Reconstruction of signal from its samples using interpolation

first-order hold is the linear interpolation. Has the following filter – inset image pg 178

7.3 Effect of undersampling (Aliasing)

Do the example with a $\cos\omega_0 t$ and conclude that undersampling turns high frequencies into low frequencies. Stroboscopic effect

7.4 Discrete-time processing of continuous time signals

First we must transform continuous to discrete (C/D conversion , Analog to digital), then process the discrete signal and lastly convert it back to continuous time (D/C conversion, Digital to Analog).

C/D conversion

it can be effectively by impulse train.

$$x_p(t) = \begin{cases} x_c(nT) & t = nT \\ 0 & \text{otherwise} \end{cases} \qquad x_d[n] = x_p(nT)$$

then

$$\begin{aligned} X_d\left(e^{j\Omega}\right) &= \sum_{k=-\infty}^{\infty} x_d[k]e^{-j\Omega k} \\ &= \sum_{k=-\infty}^{\infty} x_p(nT)e^{-j\Omega k} \\ &= \sum_{k=-\infty}^{\infty} x_c(nT)e^{-j\Omega k} \end{aligned}$$

$$= X_p\left(j\frac{\Omega}{T}\right)$$

$$\implies X_d\left(e^{j\Omega}\right) = \frac{\omega_s}{2\pi}\sum_{k=-\infty}^{\infty} X_c\left(j\frac{\Omega}{T} - jk\omega_s\right)$$

D/C Conversion

we just reverse back. we turn discrete time into impulse train and then apply a lowpass filter to get the continuous signal.

The system

give a sufficiently band limited input and sampling theorem conditions hold is LTI.

$$H_c(j\omega) = \begin{cases} H_d\left(e^{j\omega T}\right) & |\omega| < \frac{\omega_s}{2} \\ 0 & \text{otherwise} \end{cases}$$

Half-sample delay

suppose we want to do

$$y_c(t) = x_c(t - \Delta)$$

Then

$$Y_c(j\omega) = e^{-j\omega\Delta}X_c(j\omega)$$

implying that

$$H_c(j\omega) = \begin{cases} e^{-j\omega\Delta} & |\omega| < \omega_c \\ 0 & \text{otherwise} \end{cases}$$

and

$$H_d\left(e^{j\Omega}\right) = e^{-j\Omega\frac{\Delta}{T}}$$

given that $\frac{\Delta}{T}$ is an integer

$$y_d[n] = x_d\left[n - \frac{\Delta}{T}\right]$$

7.5 Sampling of discrete-time signal

Implust train signal

$$x_p[n] = \begin{cases} x(n) & N|n \\ 0 & \text{otherwise} \end{cases}$$

where N is the sampling period. THen,

$$\begin{aligned} x_p[n] &= x[n]p[n] \\ &= \sum_{k=-\infty}^{\infty} x[kN]\delta[n - kN] \end{aligned}$$

$$
\begin{aligned}
P\left(e^{j\omega}\right) &= \frac{2\pi}{N}\sum_{k=-\infty}^{\infty}\delta(\omega - k\omega_s) \\
\implies X_p\left(e^{j\omega}\right) &= \frac{1}{2\pi}\int_{2\pi} P\left(e^{j\theta}\right)X\left(e^{j(\omega-\theta)}\right)\mathrm{d}\theta \\
&= \frac{1}{N}\int_0^{2\pi} X\left(e^{j(\omega-\theta)}\right)\sum_{k=-\infty}^{\infty}\delta(\theta - k\omega_s)\,\mathrm{d}\theta \\
&= \frac{1}{N}\sum_{k=0}^{N-1}\int_0^{2\pi} X\left(e^{j(\omega-\theta)}\right)\delta(\theta - k\omega_s)\,\mathrm{d}\theta \\
&= \frac{1}{N}\sum_{k=0}^{N-1} X\left(e^{j(\omega-k\omega_s)}\right)
\end{aligned}
$$

again no aliasing if $\omega_s > 2\omega_M$.

Discrete-time decimation

it is unnecessary to keep all the zeros.

$$x_b[n] = x_p[nN] = x[nN]$$

therefore

$$
\begin{aligned}
X_b\left(e^{j\omega}\right) &= \sum_{k=-\infty}^{\infty} x_b[k]e^{-j\omega k} \\
&= \sum_{k=-\infty}^{\infty} x_p[kN]e^{-j\omega k} \\
&= \sum_{k=-\infty}^{\infty} x_p[k]e^{-j\omega/Nk} \\
&= X_p\left(e^{j\omega/N}\right)
\end{aligned}
$$

7.5.1 upsampling or interpolation

adding zeros $x_b[n] \to x_p[n] \to x[n]$ and then apply low-pass filter.